

ER Diagram for Municipal OTMS (Online Tax Management System)

Core Entities

1. Ward

Represents municipal administrative divisions.

Attributes:

- ward_id (PK)
 - ward_number
 - ward_name
 - zone_name
 - officer_name
 - contact_number
-

2. Property

Represents taxable properties within a ward.

Attributes:

- property_id (PK)
 - ward_id (FK)
 - property_number
 - property_type
 - usage_type
 - address
 - area_sqft
 - construction_year
 - occupancy_status
 - latitude
 - longitude
 - status
-

3. Owner

Represents owners of properties.

Attributes:

- owner_id (PK)
 - full_name
 - father_name
 - mobile_number
 - email
 - aadhaar_number
 - address
 - gender
 - date_of_birth
-

4. Property_Owner

Handles multiple owners for one property.

Attributes:

- property_owner_id (PK)
 - property_id (FK)
 - owner_id (FK)
 - ownership_percentage
 - ownership_type
 - start_date
 - end_date
-

5. Tax_Assessment

Stores yearly tax assessments including house tax, water tax, and sewer tax.

Attributes:

- assessment_id (PK)
- property_id (FK)
- financial_year
- annual_rental_value
- house_tax_amount
- water_tax_amount
- sewer_tax_amount
- other_charges
- tax_rate
- rebate_amount
- penalty_amount
- total_tax

- assessment_date
 - assessed_by
-

6. Tax_Payment

Stores payment records.

Attributes:

- payment_id (PK)
 - assessment_id (FK)
 - payment_date
 - payment_mode
 - transaction_reference
 - paid_amount
 - receipt_number
 - payment_status
-

7. Tax_Rate_Master

Stores tax rates by property category.

Attributes:

- tax_rate_id (PK)
 - property_type
 - usage_type
 - tax_percentage
 - effective_from
 - effective_to
-

8. Employee

Municipal staff handling operations.

Attributes:

- employee_id (PK)
- employee_name
- designation
- mobile_number
- email
- ward_id (FK)

- username
- password_hash
- role

9. Mutation

Handles ownership transfer and property mutation records.

Attributes:

- mutation_id (PK)
 - property_id (FK)
 - old_owner_id (FK)
 - new_owner_id (FK)
 - mutation_date
 - mutation_type
 - approval_status
 - approved_by
 - remarks
 - document_reference
-

Relationships

Entity 1	Relationship	Entity 2
Ward	One-to-Many	Property
Property	Many-to-Many	Owner
Property	One-to-Many	Tax_Assessment
Tax_Assessment	One-to-Many	Tax_Payment
Tax_Rate_Master	One-to-Many	Tax_Assessment
Ward	One-to-Many	Employee
Property	One-to-Many	Mutation
Owner	One-to-Many	Mutation (Old Owner)
Owner	One-to-Many	Mutation (New Owner)

ER Diagram (Mermaid Syntax)

```
erDiagram
    WARD ||--o{ PROPERTY : contains
    PROPERTY ||--o{ PROPERTY_OWNER : mapped
    OWNER ||--o{ PROPERTY_OWNER : owns
    PROPERTY ||--o{ TAX_ASSESSMENT : assessed_for
    TAX_ASSESSMENT ||--o{ TAX_PAYMENT : paid_by
    TAX_RATE_MASTER ||--o{ TAX_ASSESSMENT : applies
    WARD ||--o{ EMPLOYEE : managed_by
    PROPERTY ||--o{ MUTATION : mutation_for
    OWNER ||--o{ MUTATION : old_owner
    OWNER ||--o{ MUTATION : new_owner
```

```
WARD {
    int ward_id PK
    string ward_number
    string ward_name
    string zone_name
}
```

```
PROPERTY {
    int property_id PK
    int ward_id FK
    string property_number
    string property_type
    string usage_type
    string address
    float area_sqft
}
```

```
OWNER {
    int owner_id PK
    string full_name
    string mobile_number
    string aadhaar_number
}
```

```
PROPERTY_OWNER {
    int property_owner_id PK
    int property_id FK
    int owner_id FK
    float ownership_percentage
}
```

```
TAX_ASSESSMENT {
```

```

    int assessment_id PK
    int property_id FK
    string financial_year
    decimal house_tax_amount
    decimal water_tax_amount
    decimal sewer_tax_amount
    decimal other_charges
    decimal total_tax
    decimal rebate_amount
    decimal penalty_amount
}

TAX_PAYMENT {
    int payment_id PK
    int assessment_id FK
    date payment_date
    decimal paid_amount
    string payment_mode
}

TAX_RATE_MASTER {
    int tax_rate_id PK
    string property_type
    string usage_type
    decimal tax_percentage
}

EMPLOYEE {
    int employee_id PK
    string employee_name
    string designation
    int ward_id FK
}

MUTATION {
    int mutation_id PK
    int property_id FK
    int old_owner_id FK
    int new_owner_id FK
    date mutation_date
    string mutation_type
    string approval_status
}

```

Suggested Additional Modules

You can also extend the OTMS with:

- Advanced Mutation Workflow
- Water Tax
- Sewer Tax
- GIS Mapping
- Complaint Management
- Notice Generation
- Online Payment Gateway
- Audit Logs
- User Roles & Permissions
- Mobile App Integration

Recommended Database

- MySQL
- PostgreSQL
- SQL Server

PostgreSQL is recommended if GIS/location support is needed.